Task 3.1: Deep FCNN on Tiny ImageNet Report

February 3, 2026

1 Introduction

This report documents the implementation and analysis of a Deep Fully Connected Neural Network (FCNN) trained on a 10-class subset of the Tiny ImageNet dataset. The objective was to compare the training dynamics of Sigmoid activation versus Leaky ReLU with Batch Normalization in an 8-layer architecture.

Hardware Specification: The experiments were conducted on a single **NVIDIA RTX 4050 (6GB VRAM)**. The lightweight nature of the Deep FCNN allowed the entire batch and model to fit comfortably within the VRAM, ensuring efficient training.

2 Architecture

The network relies on **Circuit Theory** principles, utilizing depth to represent complex functions efficiently. It consists of an input layer (12,288 features) followed by 7 hidden layers.

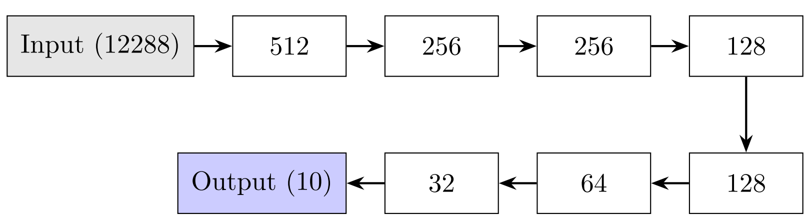

Figure 1: Schematic of the 8-Layer Deep FCNN Architecture.

3 Expectations

Prior to experimentation, we established the following expectations:

- **Vanishing Gradients:** We expect small gradient norms and the Vanishing Gradient Problem (VGP) in the Sigmoid network due to its depth.

- **ReLU + BN:** Learning should occur effectively with Leaky ReLU and Batch Normalization, as these mitigate VGP.
- **Overfitting:** As FCNNs are used on images (lacking spatial inductive bias), we expect overfitting in the absence of Dropout.
- **Complexity:** Simplifying the network architecture (reducing neurons per layer) might reduce overfitting.
- **Speed:** ReLU training speed is expected to be faster due to the elimination of computationally expensive exponential operations found in Sigmoid.

4 Experiments

4.1 Experiment A: Sigmoid (No Batch Norm)

Configuration: Activation function set to Sigmoid. No Batch Normalization applied.

Observations:

1. **Vanishing Gradients:** The gradient norms in the first layer started extremely low (0.0005), confirming the vanishing gradient problem.
2. **Performance:** The model underfitted significantly. Final Validation Accuracy reached only **26.2%**, with training loss stalling around 1.6.

4.2 Experiment B: Leaky ReLU + Batch Normalization

Configuration: Activation function set to Leaky ReLU ($\alpha = 0.01$) with Batch Normalization.

Observations:

1. **Healthy Gradient Flow:** Gradient norms remained stable (> 0.5), enabling rapid learning.
2. **Overfitting:** The model exhibited significant overfitting. Training loss dropped to **0.27**, while validation loss increased to **2.15**.
3. **Performance:** Despite overfitting, the model achieved a peak validation accuracy of **40.4%**.
4. **Regularization:** We deliberately avoided Early Stopping as it is not an orthogonal method of regularization for this analysis.

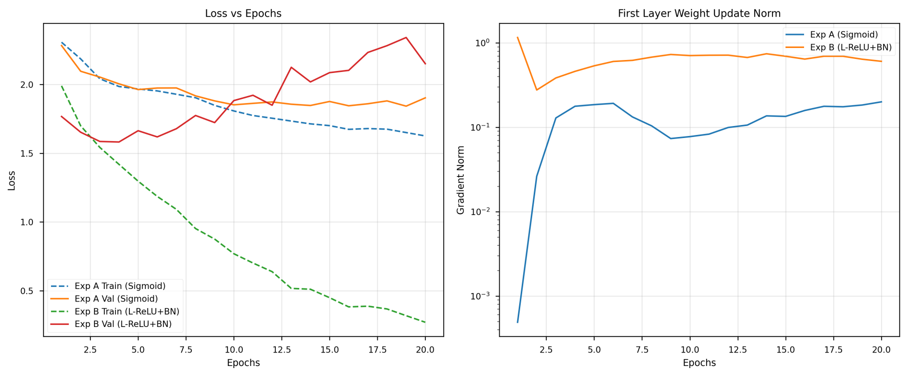

Figure 2: Training Dynamics Comparison. Left: Loss vs Epochs showing Experiment B overfitting. Right: Gradient Norms showing Sigmoid (Experiment A) vanishing gradients.

5 Visual Evidence

6 Conclusion

The experiments demonstrate that **Leaky ReLU + BN** solves the vanishing gradient problem inherent in deep Sigmoid networks. However, the deep FCNN architecture is prone to overfitting on limited image data.